Special Numbers

Year 5 Accelerated · **Homework Worksheet**

Worked Example

Square, Cube & Triangular Numbers: Find the 6th triangular number, then state whether it is also a square number.

Answer: The 6th triangular number is 1+2+3+4+5+6 = 21. Since no whole number squared equals 21 (4^2=16, 5^2=25), 21 is not a square number.

Square & Cube Basics (15 questions)

1. Write down the first six cube numbers.
2. What is 9^2?
3. What is the square root of 64?
4. What is 3^3 (3 cubed)? A) 6 B) 9 C) 18 D) 27 E) 36
5. What is the cube root of 8?
6. Is 50 a square number? Explain how you know.
7. Write down the first five triangular numbers.
8. What is the 6th triangular number? A) 15 B) 18 C) 21 D) 24 E) 28
9. What is 12^2?
10. Which of these is a cube number? A) 27 B) 30 C) 40 D) 50 E) 60
11. Select ALL that apply: which of these numbers are square numbers? A) 4 B) 12 C) 36 D) 44 E) 49
12. What is the square root of 121?
13. What is 5^3 (5 cubed)?
14. Select ALL that apply: which of these numbers are triangular numbers? A) 3 B) 5 C) 10 D) 12 E) 21
15. What is $4^2 + 2^3$? A) 22 B) 23 C) 24 D) 25 E) 26

Patterns & Classifying (15 questions)

1. A number is both a square number and an odd number, and it lies between 20 and 40. What is the number?
2. The sum of the first 4 odd numbers is 1 + 3 + 5 + 7 = 16. Use this pattern to write the sum of the first 7 odd numbers without adding them one by one.
3. What is the 9th triangular number?
4. Which of these numbers is a triangular number? A) 26 B) 27 C) 28 D) 29 E) 30
5. Select ALL that apply: which of these statements about 100 are true? A) 100 is a square number. B) 100 is a cube number. C) The square root of 100 is 10. D) 99 is a cube number. E) 100 is an odd number.
6. Find two triangular numbers that add together to make 16.
7. What is the difference between the 5th and 4th square numbers?
8. A number is a cube number greater than 100 and less than 150. What is the number? A) 64 B) 81 C) 100 D) 121 E) 125

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9. The 7th triangular number is 28. What is the 8th triangular number?

10. Select ALL that apply: which of these numbers are both square numbers and triangular numbers? A) 1 B) 4 C) 9 D) 16 E) 36

11. What is the sum of the first 6 odd numbers? What type of special number does this give?

12. Which square number, when you find its square root, gives an even number greater than 8? A) 49 B) 64 C) 81 D) 100 E) 121

13. A display of cans is stacked in a triangular pattern: 1 can in the top row, 2 in the next, 3 in the next, and so on. How many cans are in a stack with 7 rows?

14. What is $6^2 - 4^2$? A) 18 B) 19 C) 20 D) 21 E) 22

15. Explain, using an example, why a triangular number plus the next triangular number always gives a square number.

Standard / Advanced (15 questions)

1. Find the smallest square number that is greater than 200.

2. Find the largest cube number that is less than 100.

3. A square number has a square root that is itself a square number greater than 1. From the list below, find the square number whose root is 9. A) 25 B) 36 C) 49 D) 64 E) 81

4. The 12th triangular number is 78. Use this to find the 13th triangular number without listing all the terms.

5. Select ALL that apply: a number n is both a triangular number and an odd number, with $5 < n < 50$. Which of these values of n are correct? A) 10 B) 15 C) 21 D) 28 E) 45

6. Show that $8^2 + 8$ equals twice the 8th triangular number. (The nth triangular number is $n(n+1)/2$.)

7. A square garden has an area of 196 m^2. What is the length of one side of the garden?

8. Find a square number from the list below that is also one less than a multiple of 10. A) 16 B) 25 C) 36 D) 49 E) 64

9. The 9th triangular number is 45. Find the 10th and 11th triangular numbers, explaining the pattern you use.

10. A cube number is formed by cubing a whole number n. If this cube number is 512, what is n? A) 6 B) 7 C) 8 D) 9 E) 10

11. Select ALL that apply: which of these statements about square and cube numbers are true? A) Every square number is positive. B) 1 is both a square number and a cube number. C) 125 is both a square number and a cube number. D) The square of an even number is always even. E) The cube of an odd number is always odd.

12. A stack of boxes is built in triangular layers like the triangular number pattern: 1 box on top, then 2, then 3, and so on. How many boxes are needed to build a stack 9 layers deep?

13. Find the value of n if $n^2 = 196$. A) 12 B) 13 C) 14 D) 15 E) 16

14. A number is both a triangular number and a square number, and it is greater than 1 and less than 50. What is the number?

15. Using the fact that the sum of the first n odd numbers equals n^2, find the sum of the first 12 odd numbers without adding them individually.

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